

Diaphragm moment joint — hand-check sheet

Joint 10, interior (X), four IPE 160 on HSS 200x100x4. Companion to dmj_lib.m and diaphragm_joint_manual.md. Every capacity on this sheet was recomputed independently of the script and compared with it: **all values agree**. Units N, mm throughout; results shown in kN and kNm. AISC 360-16, LRFD.

1. Index of limit states

Tag	Limit state	Reference	Capacity	Gravity DCR	Seismic DCR
T1	Flange to plate weld, 3-sided	J2.4 (J2-3)	184.8 kN	0.51	0.22
T2	Beam flange gross yielding	J4.1 (J4-1)	136.5 kN	0.69	0.30
T3	Beam flange rupture at the weld	J4.2 (J4-2), U=1.0	182.0 kN	0.52	0.22
T4s	Plate yielding at the lap, strong	J4.1 (J4-1)	470.8 kN	0.20	0.09
T4w	Plate yielding at the lap, weak	J4.1 (J4-1)	470.8 kN	0.09	0.05
T5s	Plate strips in tension, strong	J4.1, [MODEL]	432.0 kN	0.22	0.09
T5w	Plate strips in tension, weak	J4.1, [MODEL]	432.0 kN	0.09	0.05
T6s	Plate strips in compression, strong	E3 (E3-2), [MODEL]	400.9 kN	0.24	0.10
T6w	Plate strips in compression, weak	E3 (E3-2), [MODEL]	424.0 kN	0.09	0.06
T7s	Collar weld to the column, strong	J2.4 (J2-3)	244.3 kN	0.01	0.19
T7w	Collar weld to the column, weak	J2.4 (J2-3)	122.2 kN	0.01	0.09
T8s	Column walls in shear, strong	J4.2 (J4-3)	216.0 kN	0.01	0.21
T8w	Column walls in shear, weak	J4.2 (J4-3)	108.0 kN	0.01	0.10
T9s	Wall local yielding, strong	cf. J10.2, [MODEL]	127.8 kN	0.02	0.36
T9w	Wall local yielding, weak	cf. J10.2, [MODEL]	90.0 kN	0.02	0.12
T10s	Wall in-plane tension, strong	[INFO], half length	180.0 kN	0.52	0.23
T10w	Wall in-plane tension, weak	[INFO], half length	90.0 kN	0.45	0.26
T11s	Wall stress interaction	von Mises, [INFO]	225.0 MPa	0.48	0.36
T11w	Wall stress interaction	von Mises, [INFO]	225.0 MPa	0.41	0.25
S1	Beam web local yielding	J10.2 (J10-3)	151.2 kN	0.10	0.03
S2	Beam web crippling at the end	J10.3 (J10-5b)	129.0 kN	0.12	0.04
S3	Bearing on the shelf plate	J7 (J7-1)	2,214.0 kN	0.01	0.00
S4	Shelf plate bending	F11.1 (F11-1)	0.664 kNm	0.93	0.29
S5	Shelf collar weld, vertical shear	J2.4 (J2-3)	733.0 kN	0.02	0.01
S6	Beam web shear	G2.1 (G2-1)	120.0 kN	0.13	0.04
M1	Beam flexure	F2.1 (F2-1)	27.878 kNm	0.58	0.25
M4	Column flexure, strong	F7.1 (F7-1)	34.229 kNm	0.02	0.23
M5	Column flexure, weak	F7.1 (F7-1)	21.089 kNm	0.01	0.09

S8 (shelf, axial plus bending) is derived from S4 in section 5. Tags marked [MODEL] or [INFO] in section 4 are engineering models, not code equations.

2. Data

2.1 Materials and sections

Item	Symbol	Value	Note
Elastic modulus	E	200 000 MPa	
Yield, all parts	Fy	250 MPa	A36
Tensile, all parts	Fu	400 MPa	A36
Electrode	FEXX	480 MPa	E70
Beam depth	h	160 mm	IPE 160
Beam flange width	bf	82 mm	
Beam flange thickness	tf	7.4 mm	
Beam web thickness	tw	5.0 mm	
Beam root radius	r	9.0 mm	
Beam plastic modulus	Zx	123 900 mm ³	
Column face, strong	D	200 mm	HSS 200x100x4
Column face, weak	B	100 mm	
Column wall	t	4.0 mm	
Column plastic modulus, strong	Zcx	152,128 mm ³	sharp corners
Column plastic modulus, weak	Zcy	93,728 mm ³	sharp corners

2.2 Plates and welds

Item	Symbol	Value	How it is obtained
Cap plate thickness	t_cap	12 mm	input
Shelf plate thickness	t_shf	12 mm	input
Lap over each beam flange	L_lap	80 mm	input; also the weld length each side
Overhang, side with no beam	w_back	50 mm	input; not used at this joint (4 beams)
Plate size, strong direction	Wx	360 mm	$D + 2 \times L_lap$
Plate size, weak direction	Wy	260 mm	$B + 2 \times L_lap$
Strip beside opening, strong	marg_s	80 mm	$(W_y - B)/2$
Strip beside opening, weak	marg_w	80 mm	$(W_x - D)/2$
Whitmore width	b_eff	174.4 mm	$\min(bf + 2 \times L_lap \tan 30^\circ, \text{plate width})$
Lever arm	z	172 mm	$h + (t_cap + t_shf)/2$
Flange weld leg	leg_fl	5 mm	min per Table J2.4, max per J2.2b = $tf - 2$
Flange weld throat	a_fl	3.53 mm	$0.707 \times \text{leg}$
Flange weld length	L_w,fl	242 mm	$2 \times L_lap + bf$, three-sided
Collar weld leg	leg_col	4 mm	matches the 4 mm wall
Collar weld throat	a_col	2.83 mm	$0.707 \times \text{leg}$
Collar weld lines, cap	n_cap	1	top face only
Collar weld lines, shelf	n_shf	2	both faces, shop welded

2.3 Seat switches (section 6 of the manual)

Switch	Value used	Alternative
e_react, reaction eccentricity	40 mm, conservative bound	25 mm realistic
share, part of V on the shelf	1.00, shelf alone	0.50, both plates hinge
bending width	bf, no dispersion	$bf + 2e$, 45 degrees
axial width for S8	Whitmore	bf

3. Demands

Quantity	How it is obtained	Gravity	Seismic
Load combination	from ETABS	1.2D+L+1.6Lr	1.2D-Ey-.3Ex
Beam 38 moment	M3 at the beam end	16.22 kNm	7.02 kNm
Beam 35 moment	M3 at the beam end	15.21 kNm	1.22 kNm
Beam 64 moment	M3 at the beam end	6.61 kNm	4.04 kNm
Beam 63 moment	M3 at the beam end	6.90 kNm	2.16 kNm
Column-top moment, strong	M3 at the column top	0.53 kNm	7.89 kNm
Column-top moment, weak	M2 at the column top	0.27 kNm	1.82 kNm
Largest reaction	-F1 on the joint, column axes	15.37 kN	4.80 kN
Column axial	-P at the column top	41.64 kN	15.10 kN
Local flange force T	max M / z	94.3 kN	40.8 kN
Local, weak direction	max M weak / z	40.1 kN	23.5 kN
Anchored force, strong	M_col strong / z	3.1 kN	45.9 kN
Anchored force, weak	M_col weak / z	1.6 kN	10.6 kN

Worked: $T = 16.22 \times 10^6 / 172 = 94\,302 \text{ N} = 94.3 \text{ kN}$. Anchored, seismic strong: $7.89 \times 10^6 / 172 = 45\,872 \text{ N} = 45.9 \text{ kN}$. The local force is largest under gravity (the beams balance, so the plate carries it across); the anchored force is largest under seismic (the beam moments add). Different governing cases, by design.

4. Capacities, with numbers substituted

Tag	Reference	Substitution	Result
T1	J2.4 (J2-3)	$0.75 \times 0.60 \times 480 \times 3.53 \times (2 \times 80 + 82)$	184.8 kN
T2	J4.1 (J4-1)	$0.90 \times 250 \times 82 \times 7.4$	136.5 kN
T3	J4.2 (J4-2), U=1.0	$0.75 \times 400 \times 82 \times 7.4$	182.0 kN
T4s	J4.1 (J4-1)	$0.90 \times 250 \times 174.4 \times 12$	470.8 kN
T4w	J4.1 (J4-1)	$0.90 \times 250 \times 174.4 \times 12$	470.8 kN
T5s	J4.1, [MODEL]	$0.90 \times 250 \times 2 \times 80 \times 12$	432.0 kN
T5w	J4.1, [MODEL]	$0.90 \times 250 \times 2 \times 80 \times 12$	432.0 kN
T6s	E3 (E3-2), [MODEL]	$KL/r=37.5, Fe=1402, Fcr=232: 0.90 \times 232 \times 2 \times 80 \times 12$	400.9 kN
T6w	E3 (E3-2), [MODEL]	$KL/r=18.8, Fe=5606, Fcr=245: 0.90 \times 245 \times 2 \times 80 \times 12$	424.0 kN
T7s	J2.4 (J2-3)	$0.75 \times 0.60 \times 480 \times 2.83 \times (1 \times 2 \times 200)$	244.3 kN
T7w	J2.4 (J2-3)	$0.75 \times 0.60 \times 480 \times 2.83 \times (1 \times 2 \times 100)$	122.2 kN
T8s	J4.2 (J4-3)	$0.90 \times 0.60 \times 250 \times 2 \times 200 \times 4$	216.0 kN
T8w	J4.2 (J4-3)	$0.90 \times 0.60 \times 250 \times 2 \times 100 \times 4$	108.0 kN
T9s	cf. J10.2, [MODEL]	$be=\min(200, 82+5 \times 12)=142: 0.90 \times 250 \times 4 \times 142$	127.8 kN
T9w	cf. J10.2, [MODEL]	$be=\min(100, 82+5 \times 12)=100: 0.90 \times 250 \times 4 \times 100$	90.0 kN
T10s	[INFO], half length	$0.90 \times 250 \times 2 \times (200/2) \times 4$	180.0 kN
T10w	[INFO], half length	$0.90 \times 250 \times 2 \times (100/2) \times 4$	90.0 kN
T11s	von Mises, [INFO]	0.90×250	225.0 MPa
T11w	von Mises, [INFO]	0.90×250	225.0 MPa
S1	J10.2 (J10-3)	$1.00 \times 250 \times 5 \times (2.5 \times 16.4 + 80)$	151.2 kN
S2	J10.3 (J10-5b)	$0.75 \times 0.40 \times 5^2 \times [1 + (4 \times 80 / 160 - 0.2) / (5 / 7.4)^{1.5}] \times \sqrt{E F_y t_f / t_w}$	129.0 kN
S3	J7 (J7-1)	$0.75 \times 1.8 \times 250 \times 82 \times 80$	2,214.0 kN
S4	F11.1 (F11-1)	$0.90 \times \min(250 \times 82 \times 12^2 / 4, 1.6 \times 250 \times 82 \times 12^2 / 6)$	0.664 kNm
S5	J2.4 (J2-3)	$0.75 \times 0.60 \times 480 \times 2.83 \times (2 \times 2 \times (200 + 100))$	733.0 kN
S6	G2.1 (G2-1)	$1.00 \times 0.60 \times 250 \times 160 \times 5$	120.0 kN
M1	F2.1 (F2-1)	$0.90 \times 123900 \times 250$	27.878 kNm
M4	F7.1 (F7-1)	$0.90 \times 152128 \times 250$	34.229 kNm
M5	F7.1 (F7-1)	$0.90 \times 93728 \times 250$	21.089 kNm

5. S8: shelf under axial force and bending

Exact plastic interaction for a rectangle, not a linear sum: axial load consumes the middle of the section, where the bending lever arm is smallest.

Step	Formula	Gravity	Seismic
Axial force	$T = \max M \text{ strong} / z$	94.3 kN	40.8 kN
Squash load of the strip	$0.90 \times 250 \times 174.4 \times 12$	470.8 kN	470.8 kN
n	T / squash	0.200	0.087
Reduced moment	$Mn \times (1 - n^2)$	0.638 kNm	0.659 kNm
Demand	$\text{share} \times V \times e$	0.615 kNm	0.192 kNm
DCR		0.96	0.29

6. Three checks to do first

1. $T = 16.22 \times 10^6 / 172 = 94\,302 \text{ N}$. If your z differs, everything scales.
2. $T2 = 0.90 \times 250 \times 82 \times 7.4 = 136\,530 \text{ N}$. Two numbers, no geometry.
3. $T8s = 0.90 \times 0.60 \times 250 \times 2 \times 200 \times 4 = 216\,000 \text{ N}$.

If these three match, the units, the resistance factors and the force are right, and the rest is the same arithmetic with different areas.

7. What is not on this sheet

Column P-M-M interaction (take it from ETABS), lateral-torsional buckling of the cantilevers, the erection cleat, the pinned short-bay connection, fatigue, fire, and seismic detailing under AISC 341 (this connection is not prequalified under AISC 358). Models marked [MODEL] and [INFO] in section 1 carry no code equation: T5, T6, T9, T10, T11 and the lever arm z . Section 8 of the manual lists them with their risk.